ENGIN OKSUZ

ROBOTICS SOFTWARE ENGINEER

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SUMMARY

- Passionate and innovative engineer who possesses a desire to constantly develop new skills
- Illustrates the true meaning of teamwork; strong communication skills and a collaborative approach to problem-solving by working closely with product managers, designers, engineers, and other stakeholders
- Highly self-motivated to work efficiently under little/no supervision

TECHNICAL SKILLS

Programming & Scripting : C++, Python , Bash
Tools and Platfroms : Docker, ROS1/ROS2, ONNX Runtime, TensorRT, NVIDIA Jetson, Ubuntu Linux
Version Control : Git, GitHub, GitLab, Bitbucket
Frameworks and Libraries : PyTorch, OpenCV, PCL, Open3D, SuperGradients, DINOv2, SAM, RF-DETR, YOLO, Autoware.AI/Universe, MAVROS
Specialized Areas : Semantic Segmentation, Object Detection, Sensor Fusion, SLAM, Collision Avoidance, Terrain Analysis
Sensor Technologies : LiDAR (Velodyne, Robosense), Cameras (Lucid, ZEDs), IMU/GPS (Apllanix, SBG), RADAR (SmartMicro)
Network Communication : CANbus, UDP/TCP, ROS topics/services
Project Management : Jira , Bitbucket , Confluence
Hardware Platforms : NVIDIA Jetson AGX/NX, Neousys Nuvo
Simulation & Testing : Gazebo, RViz
Soft Skills : Research, Technical Communication, Leadership, Team Collaboration

PROFESSIONAL WORK EXPERIENCE

Robotics Software Engineer, HAVELSAN - Robotics and Autonomous Systems, Ankara **Apr 2021 - Present**

UGV - (Unmanned Ground Vehicle) BARKAN Responsibilities:

Link : UGV BARKAN

Perception & Computer Vision:

- **Advanced Segmentation**: Advanced Segmentation: Architected PyTorch segmentation system with SuperGradients, CLAHE preprocessing and morphological operations
- **Multi-Modal Sensor Fusion**: Designed LiDAR-camera fusion with OpenCV stereo vision and PCL point cloud processing
- **Deep Learning Integration**: Implemented DINOv2 transformers and Meta SAM for feature extraction and zero-shot segmentation
- **Real-Time Object Detection**: Deployed RF-DETR models with ONNX Runtime GPU optimization achieving high-FPS detection

Sensor Fusion & Localization:

- **FAST-LIO Integration**: Led real-time SLAM implementation for UGV with IMU-LiDAR fusion
- **Multi-Sensor Processing**: Architected fusion systems for Livox LiDAR, cameras, and IMU with advanced preprocessing
- **Point Cloud Systems**: Designed 3D pipelines using PCL and Open3D for voxel filtering and outlier removal

Navigation & Path Planning:

- **Dynamic Costmap Generation**: Implemented costmap algorithms for autonomous navigation with real-time obstacle avoidance
- **Collision Avoidance**: Developed systems integrating LiDAR-camera fusion with YOLOv11 for obstacle detection and dynamic replanning

System Architecture & Monitoring:

- **Performance Monitoring**: Architected real-time CPU monitoring with C++ implementation
- **Sensor and Software Health Management**: Designed comprehensive health monitoring for sensors and ros nodes

Hardware Optimization:

- **NVIDIA Platform Deployment**: Optimized software for NVIDIA AGX/NX platforms with production-grade performance tuning
- **Hardware Optimization**: Optimized performance for Neousys Nuvo platforms using Python/C++ with production-grade tuning
- **GPU Acceleration**: Architected CUDA execution providers for ONNX models
- **ROS/ROS2 Architecture**: Designed autonomous vehicle software with extensive simulation testing

Toyota Corolla Autonomous Vehicle Responsibilities:

Link: Test Video

Sensor Integration & Data Processing:

- **Multi-Sensor Architecture**: Architected sensor fusion systems processing Lucid Cameras, Velodyne LIDARs, SmartMicro RADARs and Applanix/SBG INS data in Python/C++
- **Data Reliability Enhancement**: Implemented preprocessing algorithms and fusion techniques for enhanced reliability in autonomous navigation

Path Planning :

- **Autoware Framework Customization**: Led adaptation of Autoware path planning algorithms, implementing search-based and global planners, local planners in C++ for real-time motion planning

UAV (Unmanned Aerial Vehicle) Responsibilities:
Link: Deep Learning Based Navigation in GNSS Denied Environments
GNSS-Denied Navigation & Localization:
- **Deep Learning-Based Localization**: Architected and implemented localization algorithms using deep learning features extracted from satellite imagery with Python, achieving robust positioning in challenging environments
- **INS Integration**: Developed integration of position estimation data with INS for enhanced autopilot feedback and navigation accuracy

Computer Vision & Object Detection:
- **Real-Time Object Detection**: Designed and implemented Python-based algorithms for detecting and localizing objects from UAV camera feeds with high accuracy
- **Multi-Modal Detection Systems:** Developed detection pipelines integrating camera data with positioning for precise object localization

Data Collection & Automation:
- **Automated Data Collection**: Architected data collection processes using MAVROS and MAVSDK libraries for efficient mission execution
- **Sensor Integration:** Implemented robust systems for seamless data acquisition during autonomous flights

Indoor Positioning & Testing:
- **Indoor Positioning Systems:** Conducted comprehensive geometric flight tests with Loco Positioning System and Lighthouse indoor positioning systems using Crazyflie drone platform
- **System Validation:** Performed extensive testing and validation of positioning algorithms in controlled indoor environments
- **Link:** Crazyflie Drone Tests

Coordinator, TEKNOFEST Competitions, Istanbul — Jan 2019 - Apr 2021

- I worked as a competition coordinator in technology competitions held in 35 different fields.
- Training programs and content creation.
- Organized question and answer meeting.
- I managed website content and social media contents.

Researcher Student, HARGEM, Kayseri — Apr 2014- Apr 2016

- New generation high speed wireless communication systems research and development studies.
- Methods of Automatically Determining the Moduation Types of Communication Signals.
- DSP, FGPAs.

Electronics Engineer (Intern) , Gate Elektronik, Ankara — Jun 2015- Aug 2015

- Designed electronic circuits with necessary component selections
- Tested developed cards through prepared software.
- Implemented and trained computer vision models for object detection and image classification using TensforFlow
- Developed computer vision algorithms to detect and track object of interest in video streams using OpenCV

EDUCATION

Master of Science in Electronics Engineering — Sep 2019 - Aug 2022

Istanbul Technical University , Istanbul
- **Thesis**: Embedded AI Based Image Matching Method and Application in UAV

Bachelor's Degree in Electrical and Electronics Engineering — Sep 2012 - Jul 2017

Nuh Naci Yazgan University (100% Scholarship) , Kayseri
- **Thesis**: The Robot Can Follow The Smell
- **Degree**: Second Ranked Degree , **GPA**: 3.24 / 4.00
- **Honors**: High Honor Certificate (x2), Honor Certificate (x2)
- **Project**: TÜBİTAK 2209/A - Research Project Support Program

COURSES

- Introduction to Self-Driving Cars, *Udacity*
- Robotics Software Engineer , *Udacity*
- Indroduction to Computer Vision, *Udacity*
- C++ Programming Language Course, *Necati ERGİN*
- Advanced Python Programming, *Udemy*

HOBBIES

- Drawing
- Football
- Robotics
- Traveling